Some Notes on Convex Geometry

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Contents

The notes are written based on Daniel Hug's "A Course on Convex Geometry". The focus will be given on the volume of the convex sets and its inequalities.

1 Preliminaries

1.1 Convex sets

Definition 1.1.1. A set $A \subset \mathbb{R}^n$ is convex if for any $x, y \in \mathbb{R}^n$ and $\alpha \in [0, 1]$,

$$(1-\alpha)x + \alpha y \in A \tag{1.1}$$

Remark. The intersection of convex sets is also convex.

Definition 1.1.2. Given the points $x_1, x_2, \dots, x_k \in \mathbb{R}^n, k \in \mathbb{N}$. Any point x of the form

$$x = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_k x_k = \sum_{i=1}^{k} \alpha_i x_i \tag{1.2}$$

with $\sum_{i=1}^{k} \alpha_i = 1, \alpha_i \geq 0$ is called a convex combination of the points $x_1, x_2, \dots, x_k$.

Definition 1.1.3. Given a set $A \subset \mathbb{R}^n$. The convex hull of A, denoted by $\operatorname{conv} A$, is the intersection of all convex set containing A.

Theorem 1.1.1. For $A \subset \mathbb{R}^n$, convex hull can be defined as the set of all convex combinations of points in A, that is

$$\operatorname{conv} A := \left\{ \sum_{i=1}^{k} \alpha_i x_i : x_1, x_2, \dots, x_k \in A, \sum_{i=1}^{k} \alpha_i = 1, \alpha_i \geq 0 \right\} \tag{1.3}$$

Definition 1.1.4. The affine hull of $A \subset \mathbb{R}^n$ is defined as

$$\text{aff } A := \left\{ \sum_{i=1}^{k} \alpha_i x_i \,:\, x_1, x_2, \ldots, x_k \in A, \sum_{i=1}^{k} \alpha_i = 1, \alpha_i \in \mathbb{R} \right\}$$

Definition 1.1.5. The relative interior of $A \subset \mathbb{R}^n$ is interior of A relative to aff A, that is

$$\text{relint } A := \{x \in A : \exists \varepsilon > 0 \text{ s.t. } B_\varepsilon(x) \cap \text{aff } A \subset A\}$$

1.2 Minkowski combination

Definition 1.2.1. For the sets $A, B \subset \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$, the set

$$\alpha A + \beta B := \{\alpha a + \beta b : a \in A, b \in B\}$$

is called Minkowski combination (or linear combination) of A and B. The operation $+$ is called Minkowski addition (or vector addition).

We have special names for the following set:

- $A + B$: The sum set
- $A + x$: A translate of A by $x \in \mathbb{R}^n$
- αA: A multiple of A, $\alpha \in \mathbb{R}$
- $-A := (-1)A$: A reflection of A in the origin
- $A - B$: the difference of A and B

Definition 1.2.2. The sets A and B are homothetic if and only if $A = \alpha B + x$ or $B = \alpha A + x$ for some $x \in \mathbb{R}^n, \alpha \geq 0$.

1.3 Polytopes

Definition 1.3.1. The convex hull of finitely many points $x_1, \ldots, x_k \in \mathbb{R}^n$ is called a polytope.

Definition 1.3.2. Let P be a polytope in $\mathbb{R}^n$. A point $x \in P$ is called a vertex of P if $P \setminus \{x\}$ is convex. The set of all vertices of P is denoted by $\text{vert}\, P$.

Theorem 1.3.1. Suppose $P = \{x_1, \ldots, x_k\}$. Then x_1 is a vertex of P if and only if $x_1 \notin \text{conv}\{x_2, \ldots, x_k\}$.

Corollary 1.3.1. A polytope $P = \text{conv}\,\text{vert}\, P$.

Definition 1.3.3. The intersection of finitely many closed halfspaces is called a polyhedral set.

Theorem 1.3.2. Every polytope is a polyhedral set.

1.4 Supporting Hyperplane

Definition 1.4.1. Let $\langle \cdot, \cdot \rangle$ be the scalar product of two vectors in $\mathbb{R}^n$. The set

$$H(u, \alpha) := \{x \in \mathbb{R}^n : \langle x, u \rangle = \alpha\}, \, u \in \mathbb{R}^n \setminus \{0\}, \, \alpha \in \mathbb{R} \tag{1.4}$$

is a hyperplane of $\mathbb{R}^n$. Note that u is a normal vector of $H(u, \alpha)$. The sets

$$H^-(u, \alpha) := \{x \in \mathbb{R}^n : \langle x, u \rangle \leq \alpha\}, \quad H^+(u, \alpha) := \{x \in \mathbb{R}^n : \langle x, u \rangle \geq \alpha\}$$

denote the two closed halfspaces bounded by $H(u, \alpha)$.

Definition 1.4.2. Let $H(u, \alpha)$ be a hyperplane of $\mathbb{R}^n$ and $A \subset \mathbb{R}^n$ be closed and convex. $H(u, \alpha)$ is called a supporting hyperplane of A if $A \cap H(u, \alpha) \neq \emptyset$ and A is contained in one of the two closed halfspaces $H^-(u, \alpha)$ or $H^+(u, \alpha)$. A halfspace containing A and bounded by a supporting hyperplane of A is called a supporting halfspace of A. The set $A \cap H(u, \alpha)$ is called a support set and any $x \in A \cap H(u, \alpha)$ is a supporting point.

Theorem 1.4.1 (Support Theorem)**.** For closed convex set $A \subset \mathbb{R}^n$, if x_0 is a point on the boundary of A, then there exists a supporting hyperplane containing x_0.

1.5 Convex Function

Definition 1.5.1. A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if and only if

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$$

for $x, y \in \mathbb{R}^n$ and $\alpha \in [0, 1]$.

Remark. f is concave if $-f$ is convex.

1.5.1 Support Function

Definition 1.5.2. Let $A \subset \mathbb{R}^n$ be nonempty and convex. The support function $h_A : \mathbb{R}^n \to \mathbb{R}$ is given by

$$h_A(u) := \sup \{\langle x, u\rangle : x \in A\}$$

Remark. A is contained in the closed halfspace $H^-(u, h_A(u))$ and there is at least one point of A in $H(u, h_A(u))$.

Theorem 1.5.1. For nonempty convex set $A, B \subset \mathbb{R}^n$. The following properties hold:

(a) $h_A(\alpha u) = \alpha h_A(u)$, $\alpha \geq 0$, $u \in \mathbb{R}^n$.

(b) $h_A(u + v) \leq h_A(u) + h_A(v)$, $u, v \in \mathbb{R}^n$.

(c) $h_{\alpha A+\beta B} = \alpha h_A + \beta h_B$.

(d) $h_{-A}(u) = h_A(-u)$.

(e) $A \subset B \implies h_A \leq h_B$.

(f) h_A is finite if and only if A is bounded.

From this section onwards, we may consider h_A as a function on the Euclidean unit sphere $\mathbb{S}^{n-1}$, where $\mathbb{S}^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}$ and $\|\cdot\|$ is Euclidean norm.

1.6 Volume and Surface Area

For a convex body K, we denote $K(u), u \in \mathbb{S}^{n-1}$ to be the support set of K and $u^\perp$ be the orthogonal complement of u. Also, we consider $K(u)|u^\perp$ as the orthogonal projection of $K(u)$ to $u^\perp$.

Definition 1.6.1 (Volume and surface area of polytopes)**.** Let P be a polytope in $\mathbb{R}^n$. For $n = 1$, that is $P = [a, b]$ with $a \leq b$, we define

$$V_1 := b - a, \quad F_1 := 2$$

For $n \geq 2$, we define

$$V_n(P) := \begin{cases} \dfrac{1}{n} \displaystyle\sum_{u \in \mathbb{S}^{n-1}} h_P(u) V_{n-1}(P(u)|u^\perp), & \text{if} \dim P \geq n - 1 \\ 0, & \text{if} \dim P \leq n - 2 \end{cases}$$

and

$$F_n(P) := \begin{cases} \sum\limits_{u \in \mathbb{S}^{n-1}} V_{n-1}(P(u)|u^{\perp}), & \text{if} \dim P \geq n-1 \\ 0, & \text{if} \dim P \leq n-2 \end{cases}$$

Theorem 1.6.1. The volume V_n and surface area F_n of polytopes P and Q have the following properties:

(a) V_n is the Lebesgue measure of P, i.e. $V_n(P) = \lambda_n(P)$.

(b) V_n and F_n are invariant with respect to rigid motions

(c) $V_n(\alpha P) = \alpha^n V_n(P)$ and $F(\alpha P) = \alpha^{n-1}(P)$ for $\alpha > 0$.

(d) $V_n(P) = 0$ if and only if $\dim P \leq n-1$.

(e) If $P \subset Q$, then $V_n(P) \leq V_n(Q)$ and $F_n(P) \leq F_n(Q)$.

Proof. We will use induction on n. When $n = 1$, $V_1 = b - a$ is the Lebesgue measure on $\mathbb{R}$. Let $n \leq 2$ and (a) holds for $n-1$. By definition, $V_n(P) = F_n(P) = 0$ if $\dim P \leq n-2$. For $\dim P = n-1$, there are two support sets of P which are facets (support sets of dimension $n-1$), namely $P = P(u_0)$ and $P = P(-u_0)$ where u_0 is a normal vector to P. Then since $V_{n-1}(P(u_0)|u_0^{\perp}) = V_{n-1}(P(-u_0)|u_0^{\perp})$ and $h_P(u_0) = -h_P(-u_0)$, we obtain $V_n(P) = 0$ and $F(P) = 2V_{n-1}(P(u_0)|u_0^{\perp})$.

Consider P with $\dim P = n$. Suppose $P(u_1), \ldots, P(u_k)$ are the facets of P. If $0 \in \operatorname{int} P$, then

$$\begin{aligned} V_n(P) &= \frac{1}{n} \sum_{i=1}^{k} h_P(u_i) V_{n-1}(P(u_i)|u_i^{\perp}) \\ &= \frac{1}{n} \sum_{i=1}^{k} h_P(u_i) \lambda_{n-1}(P(u_i)|u_i^{\perp}) \quad \text{by induction hypothesis} \end{aligned}$$

We use a fact from geometry and calculus for determining the volume of pyramid $\operatorname{conv}(\{0\} \cup P(u_i))$, which is $\lambda_n(\operatorname{conv}(\{0\} \cup P(u_i))) = \frac{1}{n} \cdot \lambda_{n-1}(P(u_i)|u_i^{\perp}) \cdot h_P(u_i)$. Hence, we obtain

$$V_n(P) = \sum_{i=1}^{k} \lambda_n(\operatorname{conv}(\{0\} \cup P(u_i))) = \lambda_n(P)$$

since the pyramids have pairwise disjoint interiors. Now we still assume that $0 \in \operatorname{int} P$. If $t \in \mathbb{R}^n$ and $\|t\|$ is small enough so that $-t \in \operatorname{int} P$. Then $0 \in \operatorname{int}(P+t)$. Using the translation invariance of λ_n we get

$$V_n(P+t) = \lambda_n(P+t) = \lambda_n(P) = V_n(P).$$

Next, since we have $h_{P+t}(u_i) = h_P(u_i) + \langle t, u_i \rangle$. Observe that $(P+t)(u_i) = P(u_i) + t$, we get

$$V_{n-1}((P+t)(u_i)|u_i^{\perp}) = V_{n-1}(P(u_i)|u_i^{\perp})$$

and

$$\begin{aligned} V_n(P+t) &= \frac{1}{n} \sum_{i=1}^{k} (h_P(u_i) + \langle t, u_i \rangle) V_{n-1}(P(u_i)|u_i^{\perp}) \\ &= V_n(P) + \frac{1}{n} \left\langle t, \sum_{i=1}^{k} u_i V_{n-1}(P(u_i)|u_i^{\perp}) \right\rangle \end{aligned}$$

Since $V_n(P) = V_n(P+t)$ when $\|t\|$ is small enough,

$$\left\langle t, \sum_{i=1}^{k} u_i V_{n-1}(P(u_i)|u_i^{\perp}) \right\rangle = 0. \tag{1.5}$$

Hence, for all $t \in \mathbb{R}^n$, $V_n(P) = V_n(P+t)$. Now let P be arbitrary, then there exists $t_0 \in \mathbb{R}^n$ such that $0 \in \operatorname{int}(P+t_0)$ and hence

$$V_n(P) = V_n(P+t_0) = \lambda_n(P+t_0) = \lambda(P)$$

which completes the proof of (a). For (b), (c), (d) and the first part of (e) follow from the properties of Lebesgue measure. We want to show $F_n(P) \leq F_n(Q)$ for $P \subset Q$. Assume that $\dim Q = n$. Suppose T is a polytope with facets $T(u_1), \dots, T(u_k)$. Then from equation (1.5) we replace P with T and get

$$\begin{aligned}
&\left\langle t, \sum_{i=1}^{k} u_i V_{n-1}(T(u_i)|u_i^{\perp}) \right\rangle = 0 \implies \sum_{i=1}^{k} u_i V_{n-1}(T(u_i)|u_i^{\perp}) = 0 \\
&\implies u_i V_{n-1}(T(u_i)|u_i^{\perp}) = -\sum_{j \neq i} u_j V_{n-1}(T(u_j)|u_j^{\perp}) \\
&\implies \left\| u_i V_{n-1}(T(u_i)|u_i^{\perp}) \right\| \leq \left\| \sum_{j \neq i} u_j V_{n-1}(T(u_j)|u_j^{\perp}) \right\| \\
&\implies V_{n-1}(T(u_i)|u_i^{\perp}) \leq \sum_{j \neq i} V_{n-1}(T(u_j)|u_j^{\perp}) \quad \because \|u_j\| = 1 \ \forall 1 \leq j \leq k
\end{aligned}$$

This implies $F_n(Q \cap H) \leq F_n(Q)$ for any closed halfspace $H \subset \mathbb{R}^n$. Since P is a finite intersection of halfspaces, we have $F_n(P) \leq F_n(Q)$ by successive truncation. $\square$

Now we write V and F instead of V_n and F_n respectively. We call the nonempty compact convex sets as convex bodies.

Theorem 1.6.2 (Approximation of Convex Bodies by Polytopes)**.** Let K be a convex body in $\mathbb{R}^n$. We define

$$V_+(K) := \inf_{P \supset K} V(P), \quad V_-(K) := \sup_{P \subset K} V(P)$$

and

$$F_+(K) := \inf_{P \supset K} F(P), \quad F_-(K) := \sup_{P \subset K} F(P)$$

where P is polytopes in R^n. If $V(K)$ and $F(K)$ denote the volume of K and the surface area of K respectively. Then for any convex body K,

$$\begin{aligned}
V_+(K) &= V_-(K) =: V(K) \\
F_+(K) &= F_-(K) =: F(K)
\end{aligned}$$

and V is the Lebesgue measure of K.

Proof. Let K be arbitrary convex body, from Theorem 1.6.1 (e) we have

$$V_-(K) \leq V_+(K) \text{ and } F_-(K) \leq F_+(K)$$

Since $V_-(K), V_+(K), F_-(K), F_+(K)$ are translation invariant, we assume $0 \in \operatorname{relint} K$. We will use the following lemma to find a suitable polytope:

Lemma 1.1. If $0 \in \operatorname{relint} K$, then there exists a polytope P which satisfies $P \subset K \subset (1+\varepsilon)P$.

Sketch of proof. Assume without loss of generality that $\dim K = n$. Hence $0 \in \operatorname{int} K$. Let $A(u, h_K(u))$ be the complement of $H^-(u, h_K(u))$ where $u \in \mathbb{S}^{n-1}$. Observe that the boundary points of $K + \varepsilon K$ are outside K. So, we can separate the boundary points by supporting hyperplanes, that is, there exist $u_1, \ldots, u_m \in \mathbb{S}^{n-1}$ such that

$$\operatorname{bd}(K + \varepsilon K) \subset \bigcup_{i=1}^{m} A(u_i, h_K(u_i)).$$

Then we can construct a polytope $P' = \bigcap_{i=1}^{m} H^-(u_i, h_K(u_i))$ so that $K \subset P' \subset (1+\varepsilon)K$. Take $P = \frac{1}{1+\varepsilon}P'$ then we can get $P \subset K \subset (1+\varepsilon)P$. ☐

So, for $\varepsilon \in (0,1)$, we obtain a P with $P \subset K \subset (1+\varepsilon)P$ from the lemma above. From Theorem 1.6.1 we have

$$V(P) \leq V_-(K) \leq V_+(K) \leq V((1+\varepsilon)P) = (1+\varepsilon)^n V(P)$$

and

$$F(P) \leq F_-(K) \leq F_+(K) \leq F((1+\varepsilon)P) = (1+\varepsilon)^{n-1} F(P).$$

As $\varepsilon \to 0$, we can get $V_-(K) = V_+(K)$, $F_-(K) = F_+(K)$ and hence the result follows. ☐

Remark. $V(K)$ is the Lebesgue measure of K.

2 Some Inequalities for Volume of Convex Sets

2.1 Brunn-Minkowski Theorem

Theorem 2.1.1 (Brunn-Minkowski Inequality)**.** Let $\mathcal{K}^n$ be the set of all convex bodies in $\mathbb{R}^n$. For any $A, B \in \mathcal{K}^n$ and for $\alpha \in [0,1]$,

$$V_n\left((1-\alpha)A + \alpha B\right)^{1/n} \geq (1-\alpha)V_n(A)^{1/n} + \alpha V_n(B)^{1/n} \tag{2.1}$$

Equality holds for some $\alpha \in (0,1)$ iff A and B either lie in parallel hyperplanes or are homothetic.

2.1.1 Proof

We present the proof produced by Kneser and Süss. The following lemma is used in the proof.

Lemma 2.1. For $\alpha \in (0,1)$ and $r, s, t > 0$,

$$\left((1-\alpha)r^t + \alpha s^t\right)^{\frac{1}{t}} \left(\frac{1-\alpha}{r} + \frac{\alpha}{s}\right) \geq 1 \tag{2.2}$$

with equality iff $r = s$.

Proof. Since $\ln x$ is strictly concave on $(0, \infty)$. We have

$$\begin{aligned}
\ln\left[\left((1-\alpha)r^t + \alpha s^t\right)^{\frac{1}{t}} \left(\frac{1-\alpha}{r} + \frac{\alpha}{s}\right)\right] &= \frac{1}{t}\ln\left((1-\alpha)r^t + \alpha s^t\right) + \ln\left(\frac{1-\alpha}{r} + \frac{\alpha}{s}\right) \\
&\geq \frac{1}{t}\left((1-\alpha)\ln r^t + \alpha \ln s^t\right) + (1-\alpha)\ln\frac{1}{r} + \alpha\ln\frac{1}{s} \\
&= (1-\alpha)\ln r + \alpha \ln s - (1-\alpha)\ln r - \alpha \ln s \\
&= 0
\end{aligned}$$

Since $\ln x$ is strictly increasing, (2.2) follows. Note that

$$\ln\left[\left((1-\alpha)r^t+\alpha s^t\right)^{\frac{1}{t}}\left(\frac{1-\alpha}{r}+\frac{\alpha}{s}\right)\right]=0 \iff r=s$$

$\square$

Proof. Denote

$$K_\alpha := (1-\alpha)A+\alpha B$$

There are few cases for A and B.

- A and B lie in parallel hyperplanes or are homothetic.
- $\dim A < n$ and $\dim B < n$.
- $\dim A < n$ and $\dim B = n$.
- $\dim A = \dim B = n$.

First case. If A and B lie in parallel hyperplanes. Then K_α lies in a hyperplane. Hence $V_n(A) = V_n(B) = 0$ and $V_n(K_\alpha) = 0$ for $\alpha \in [0,1]$. Thus equality holds in (2.1). On the other hand, equality holds trivially if A and B are homothetic.

Second case. Suppose $\dim A < n$ and $\dim B < n$. Then $V_n(A) = V_n(B) = 0$ and (2.1) holds since $V_n(K_\alpha) \geq 0$. With this condition, if $V_n(K_\alpha) = 0$, then K_α lies in a hyperplane. Since $K_\alpha = (1-\alpha)A + \alpha B$, A and B lie in parallel hyperplanes.

Third case. Suppose $\dim A < n$ and $\dim B = n$. Then $V_n(A) = 0$. We use the fact that for any $x \in A$,

$$(1-\alpha)x+\alpha B \subset (1-\alpha)A+\alpha B = K_\alpha$$

Hence,

$$\begin{aligned} V_n\left((1-\alpha)x+\alpha B\right) \leq V_n(K_\alpha) &\implies V_n(\alpha B) \leq V_n(K_\alpha)\\ &\implies \alpha^n V_n(B) \leq V_n(K_\alpha)\\ &\implies \alpha V_n(B)^{1/n} \leq V_n(K_\alpha)^{1/n} \end{aligned}$$

Note that equality holds iff $A = \{x\}$. In this case A and B are homothetic.

Fourth case. Suppose $\dim A = \dim B = n$. We can assume that $V_n(A) = V_n(B) = 1$ and prove

$$V_n((1-\alpha)A+\alpha B)^{1/n} \geq 1 \tag{2.3}$$

Then for any $A, B \in \mathcal{K}^n$ and $\alpha \in [0,1]$, we can take

$$A_* = \frac{A}{V_n(A)^{1/n}}, \quad B_* = \frac{B}{V_n(B)^{1/n}}, \quad \alpha_* = \frac{\alpha V_n(B)^{1/n}}{(1-\alpha)V_n(A)^{1/n}+\alpha V_n(B)^{1/n}}$$

Since $V_n(A_*) = V_n(B_*)$ and $\alpha_* \in [0,1]$, if (2.3) holds then we obtain

$$\begin{aligned} V_n\left(\left(\frac{(1-\alpha)A}{(1-\alpha)V_n(A)^{1/n}+\alpha V_n(B)^{1/n}}\right)+\frac{\alpha B}{(1-\alpha)V_n(A)^{1/n}+\alpha V_n(B)^{1/n}}\right) &\geq 1\\ \frac{1}{[(1-\alpha)V_n(A)^{1/n}+\alpha V_n(B)^{1/n}]^n}V_n((1-\alpha)A+\alpha B) &\geq 1 \end{aligned}$$

This implies

$$V_n((1-\alpha)A+\alpha B)^{1/n} \geq (1-\alpha)V_n(A)^{1/n} + \alpha V_n(B)^{1/n}$$

We will prove (2.3) by induction on n. For $n = 1$, equality always holds since any interval A, B can be expressed as $A = \alpha B + x$ for some $\alpha \geq 0$ and $x \in \mathbb{R}$, which means they are homothetic.

So, we assume that $n \geq 2$ and (2.3) holds in dimension $n-1$. Choose $u \in \mathbb{S}^{n-1}$ and consider $H(u, \alpha), \alpha \in \mathbb{R}$. By Fubini's theorem,

$$V_n(A \cap H^-(u,\alpha)) = \int_{-h_A(-u)}^{\alpha} V_{n-1}(A \cap H(u,\eta))\, d\eta$$

Since $V_{n-1}(A \cap H(u,\eta))$ is positive and continuous on $(-h_A(-u), h_A(u))$, the function

$$f : [-h_A(-u), h_A(u)] \to [0,1], \quad f(\alpha) = V_n(A \cap H^-(u,\alpha))$$

is differentiable on $(-h_A(-u), h_A(u))$ and $f'(\alpha) = V_{n-1}(A \cap H(u,\alpha))$. Note that f is strictly increasing, onto and continuous on $[-h_A(-u), h_A(u)]$. Hence, f is invertible and its inverse $g_A : [0,1] \to [-h_A(-u), h_A(u)]$ which is also strictly increasing and continuous, satisfies $g_A(0) = -h_A(-u), g_A(1) = h_A(u)$ and

$$g_A'(\tau) = \frac{1}{f'(g_A(\tau))} = \frac{1}{V_{n-1}(A \cap H(u, g_A(\tau)))}, \quad \tau \in (0,1)$$

Similarly, for convex body B we can get a function $g_B : [0,1] \to [-h_B(-u), h_B(u)]$ satisfying $g_B(0) = -h_B(-u), g_B(1) = h_B(u)$ and

$$g_B'(\tau) = \frac{1}{V_{n-1}(B \cap H(u, g_B(\tau)))}, \quad \tau \in (0,1)$$

For any $\alpha, \tau \in [0,1]$, consider any point $M \in (1-\alpha)(A \cap H(u, g_A(\tau))) + \alpha(B \cap H(u, g_B(\tau)))$. Notice that M divides the distance PQ in the ratio $1-\alpha : \alpha$, where $P \in A \cap H(u, g_A(\tau))$ and $Q \in B \cap H(u, g_B(\tau))$. Hence, M lies in $(1-\alpha)A + \alpha B$ and also in $H(u, (1-\alpha)g_A(\tau) + \alpha g_B(\tau))$. So,

$$\begin{aligned}&(1-\alpha)(A \cap H(u, g_A(\tau))) + \alpha(B \cap H(u, g_B(\tau))) \\ &\quad \subset ((1-\alpha)A + \alpha B) \cap H(u, (1-\alpha)g_A(\tau) + \alpha g_B(\tau))\end{aligned}$$

We compute the volume of $K_\alpha = (1-\alpha)A + \alpha B$ using the induction hypothesis. Let $c_\alpha = (1-\alpha)(-h_A(-u)) + \alpha(-h_B(-u))$ and $C_\alpha = (1-\alpha)h_A(u) + \alpha h_B(u)$. Then

$$V_n(K_\alpha) = \int_{c_\alpha}^{C_\alpha} V_{n-1}(K_\alpha \cap H(u,\eta))\, d\eta$$

Using a subtitution by the map $\eta : [0,1] \to [c_\alpha, C_\alpha]$ defined as $\eta(\tau) = (1-\alpha)g_A(\tau) + \alpha g_B(\tau)$. Note that this mapping satisfies $\eta(0) = (1-\alpha)(-h_A(-u)) + \alpha(-h_B(-u)) = c_\alpha$ and $\eta(1) = (1-\alpha)h_A(u) + \alpha h_B(u) = C_\alpha$. This implies

$$d\eta = (1-\alpha)g_A'(\tau) + \alpha g_B'(\tau)\, d\tau$$

and

$$
\begin{aligned}
V_n(K_\alpha) &= \int_0^1 V_{n-1}(K_\alpha \cap H(u,(1-\alpha)g_A(\tau)+\alpha g_B(\tau))) \cdot ((1-\alpha)g_A'(\tau)+\alpha g_B'(\tau))\, d\tau \\
&\geq \int_0^1 V_{n-1}[(1-\alpha)(A\cap H(u,g_A(\tau)))+\alpha(B\cap H(u,g_B(\tau)))] \\
&\qquad \times \left(\frac{1-\alpha}{V_{n-1}(A\cap H(u,g_A(\tau)))}+\frac{\alpha}{V_{n-1}(B\cap H(u,g_B(\tau)))}\right) d\tau \\
&\geq \int_0^1 \left((1-\alpha)V_{n-1}[A\cap H(u,g_A(\tau))]^{\frac{1}{n-1}}+\alpha V_{n-1}[B\cap H(u,g_B(\tau))]^{\frac{1}{n-1}}\right)^{n-1} \\
&\qquad \times \left(\frac{1-\alpha}{V_{n-1}(A\cap H(u,g_A(\tau)))}+\frac{\alpha}{V_{n-1}(B\cap H(u,g_B(\tau)))}\right) d\tau
\end{aligned}
$$

Choose

$$
r = V_{n-1}[A\cap H(u,g_A(\tau))], \quad s = V_{n-1}[B\cap H(u,g_B(\tau))], \quad t=\frac{1}{n-1}
$$

Then $V_n(K_\alpha)\geq 1$ by Lemma 2.2 and hence (2.1) follows. For equality, suppose $V_n(K_\alpha)=1$, then $V_{n-1}[A\cap H(u,g_A(\tau))]=V_{n-1}[B\cap H(u,g_B(\tau))]$ for $\tau\in[0,1]$ by Lemma 2.2. Therefore, for $\tau\in(0,1)$,

$$
g_A'(\tau)=\frac{1}{V_{n-1}[A\cap H(u,g_A(\tau))]}=\frac{1}{V_{n-1}[B\cap H(u,g_B(\tau))]}=g_B'(\tau)
$$

and $g_A(\tau)-g_B(\tau)$ is a constant on $[0,1]$. We may assume that the centroids of A and B are at the origin where the centroid of set A is defined as the point $c\in\mathbb{R}^n$ for which

$$
\langle c,u\rangle = \frac{1}{V_n(A)}\int_{x\in A}\langle x,u\rangle\, dx.
$$

So, we obtain

$$
\begin{aligned}
0 &= \int_{x\in A}\langle x,u\rangle\, dx \\
&= \int_{-h_A(-u)}^{h_A(u)} \eta V_{n-1}[A\cap H(u,\eta)]\, d\eta \\
&= \int_{-h_A(-u)}^{h_A(u)} \eta f'(\eta)\, d\eta \\
&= \int_0^1 g_A(\tau)\, d\tau \quad \text{by letting } \eta = g_A(\tau)
\end{aligned}
$$

Similarly, we get $0=\int_0^1 g_B(\tau)\, d\tau$. Therefore,

$$
\int_0^1 (g_A(\tau)-g_B(\tau))\, d\tau = 0 \implies g_A=g_B
$$

In particular,

$$
h_A(u)=g_A(1)=g_B(1)=h_B(u)
$$

Since u is arbitrary, we have $h_A=h_B \implies A=B$. For the converse, it is clear to see that $A=B \implies V(K_\alpha)=1$.

□

Corollary 2.1.1. The following inequalities are equivalent:

(a) $V_n(A+B)^{1/n} \geq V_n(A)^{1/n} + V_n(B)^{1/n}$

(b) $V_n(sA+tB)^{1/n} \geq sV_n(A)^{1/n} + tV_n(B)^{1/n}$ for all $s, t \geq 0$

(c) $V_n\left((1-\alpha)A + \alpha B\right) \geq V_n(A)^{1-\alpha} V_n(B)^{\alpha}$ for all $\alpha \in [0,1]$

2.2 Isoperimetric Inequality

Let B^n be a unit ball with center at origin in $\mathbb{R}^n$.

Theorem 2.2.1 (Isoperimetric Inequality)**.** If A is a convex body in $\mathbb{R}^n$, then

$$\left(\frac{F_n(A)}{F_n(B^n)}\right)^n \geq \left(\frac{V_n(A)}{V_n(B^n)}\right)^{n-1}$$

Theorem 2.2.2 (Simplified Minkowski-Steiner formula)**.** The surface area of convex body A in $\mathbb{R}^n$ is defined by

$$F_n(A) = \lim_{\varepsilon \to 0} \frac{V_n(A + \varepsilon B^n) - V_n(A)}{\varepsilon}$$

Example. If $A = B^n$, then $F_n(B^n) = nV(B^n)$.

2.2.1 Proof

Proof. Using Minkowski-Steiner formula and Brunn-Minkowski Inequality, we get

$$\begin{aligned} F_n(A) &= \lim_{\varepsilon \to 0} \frac{V_n(A + \varepsilon B^n) - V_n(A)}{\varepsilon} \\ &\geq \lim_{\varepsilon \to 0} \frac{\left(V_n(A)^{1/n} + \varepsilon V_n(B^n)^{1/n}\right)^n - V_n(A)}{\varepsilon} \\ &= nV_n(A)^{\frac{n-1}{n}} V_n(B^n)^{1/n} \end{aligned}$$

Since $F_n(B^n) = nV(B^n)$, it is written as

$$\left(\frac{F_n(A)}{F_n(B^n)}\right)^n \geq \left(\frac{V_n(A)}{V_n(B^n)}\right)^{n-1}$$

□